`bispectrum`: Selective G-Bispectra Made Practical

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Abstract

Many machine-learning tasks are invariant under group actions: image classification under planar rotation $\mathrm{SO}(2)$, signal classification under translation, and spherical-image classification under 3D rotations $\mathrm{SO}(3)$. The G-bispectrum is a principled *complete* invariant (it preserves all signal information up to the group action) but its deployment has been hampered by quadratic cost and a patchwork of group-specific implementations. We present `bispectrum`, an open-source, fully unit-tested PyTorch library that computes *selective* G-bispectral invariants for seven different group/domain pairs as drop-in differentiable `nn.Modules`. For finite groups, selectivity reduces the invariant size from $O(|G|^2)$ for the full bispectrum to $O(|G|)$ while remaining complete on generic signals. For $\mathrm{SO}(3)$ on S^2, we introduce an augmented selective construction Φ_{sel} with $\Theta(L^2)$ coefficients at band-limit L (matching the orbit-space lower bound $(L+1)^2 - 3$ up to a constant, vs. $O(L^3)$ for the full bispectrum); a formal completeness proof for the continuous case is left to future work. Gradient-descent reconstruction from Φ_{sel} produces visually aligned reconstructions with relative image-space residuals in the 0.1–0.3 range across our digit sweep (Section 6.2). We benchmark the library on three datasets across three group/domain pairs (PCam under C_8, OrganMNIST3D under the octahedral group, and Spherical MNIST under $\mathrm{SO}(3)$) comparing against norm pooling, gated pooling, Fourier-ELU pooling, max pooling, and (non-equivariant) augmented CNN baselines. The library delivers near-exact rotation invariance (test-set rotation-accuracy std $\sigma_{\mathrm{rot}} < 10^{-3}$) and a data-efficiency advantage over incomplete alternatives in the low-data, moderate-capacity regime; the selective forward pass runs in sub-millisecond time on GPU for finite groups. Code is available at `https://anonymous.4open.science/r/bispectrum/`.

1 Introduction

Many machine-learning tasks are invariant under the action of a symmetry group G: histopathology classification under planar rotations C_n, signal classification under translations, volumetric organ classification under cube rotations O, and spherical-image classification under 3D rotations $\mathrm{SO}(3)$. Equivariant neural networks [Cohen and Welling, 2016, Weiler and Cesa, 2019] enforce these symmetries by construction in their feature maps, but at the point where a prediction must be made any task that does not depend on the group action — classification, regression, detection — still has to map the equivariant features to an *invariant* representation.

The *G-bispectrum* is a complex-valued vector of triple-correlation coefficients of the group Fourier transform. It has three properties that matter for ML: (i) **completeness** — it preserves all signal information up to the group action (the map is one-to-one on group orbits); (ii) **inversion** — the signal can in principle be recovered from the invariant up to its orbit; and (iii) **selectivity** — a small $O(|G|)$ index subset of the $O(|G|^2)$ full bispectrum still suffices for completeness on generic signals [Mataigne et al., 2024].

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Existing invariant pooling layers — norm pooling, gate-based pooling, max/sum aggregation [Weiler and Cesa, 2019] — are simple but *incomplete*: they discard information distinguishing signals in different orbits. The bispectrum is a principled alternative [Giannakis, 1989, Kakarala, 2012, Sanborn et al., 2023, Mataigne et al., 2024], but existing constructions are scattered across theory papers and domain-specific implementations, and the selective template of Mataigne et al. [2024] is specific to finite groups; the continuous case of $\mathrm{SO}(3)$ on S^2 has remained open.

This paper presents `bispectrum`, a PyTorch [Paszke et al., 2019] library for computing bispectral invariants (and, where available, their inverses) on the C_n, D_n, $\mathbb{T}^d$, disk, S^2, and octahedral domains as drop-in `nn.Modules`.

Contributions.

1. **Library.** A PyTorch library implementing selective G-bispectra for *seven group/domain pairs* — C_n, $\mathrm{SO}(2)$ on S^1, D_n, $\mathbb{T}^d$, $\mathrm{SO}(2)$ on the disk, $\mathrm{SO}(3)$ on S^2, and the chiral octahedral group O on voxel grids — as drop-in differentiable `nn.Modules` with a uniform API, selective $O(|G|)$ default, precomputed Clebsch–Gordan buffers, sub-millisecond GPU forward passes for finite groups, and a fully unit-tested codebase (96.8% line coverage).
2. **Augmented selective $\mathrm{SO}(3)$ invariant on S^2.** We construct Φ_{sel} (Sec. 6), which extends the breadth-first-search (BFS) finite-group template of Mataigne et al. [2024] to the continuous case by augmenting a selective scalar bispectrum with degree-4 *CG-power* entries $P_{\ell_1,\ell_2,\ell} = \|(\hat{f}_{\ell_1} \otimes \hat{f}_{\ell_2})|_\ell\|^2$ that restore rank on real-valued signals. At band-limit L, Φ_{sel} uses $\Theta(L^2)$ scalar coefficients (matching the orbit-space lower bound $(L+1)^2 - 3$ up to a constant) versus $O(L^3)$ for the full bispectrum. Gradient-descent reconstruction from Φ_{sel} produces visually aligned copies across our digit sweep; a formal completeness proof for the continuous case is left to future work.
3. **Benchmarks.** PCam under C_8, OrganMNIST3D under the octahedral group O, and Spherical MNIST under $\mathrm{SO}(3)$, against norm pooling, gated pooling, Fourier-ELU pooling, max pooling, and (non-equivariant) augmented-CNN baselines. Bispectral pooling delivers near-exact rotation invariance (test-set $\sigma_{\mathrm{rot}} < 10^{-3}$) and a consistent data-efficiency advantage over incomplete alternatives in the low-data, moderate-capacity regime, converging to them at high capacity.

2 Mathematical Background

Notation. For a group G with associative product, a *representation* ρ assigns each $g \in G$ a unitary matrix $\rho(g)$ with $\rho(gh) = \rho(g)\rho(h)$; an *irreducible representation* (*irrep*) has no nontrivial invariant subspace, and the *dual* $\hat{G}$ is the set of equivalence classes of irreps (for finite G, $|\hat{G}| = O(|G|)$). A G-action on a domain X, $(g, x) \mapsto g \cdot x$, induces $(f \circ g)(x) := f(g \cdot x)$; two signals are in the same *orbit* iff related by some $g \in G$, and an *invariant* of f is a function constant on each orbit. For compact G, $d\mu$ denotes the normalised Haar measure (counting measure for finite G). "Lebesgue-almost-every f" means outside a measure-zero algebraic subvariety. The groups and domains we target are $C_n, D_n, \mathbb{T}^d, \mathrm{SO}(2)$ on the disk, $\mathrm{SO}(3)$ on S^2, and the chiral octahedral group O on 3D voxel grids.

2.1 Equivariant features and invariant maps

A neural network layer Φ is *G-equivariant* if its output transforms predictably under the group action: $[\pi_{\mathrm{out}}(g) \circ \Phi](f) = \Phi(\pi_{\mathrm{in}}(g)f)$ for all $g \in G$ [Cohen and Welling, 2016, Weiler and Cesa, 2019]. Here f is an input signal (a function on the domain X), $\pi_{\mathrm{in}}, \pi_{\mathrm{out}}$ are the (representations of the) G-actions on the input and output feature spaces of Φ, and the composition $\pi_{\mathrm{out}}(g) \circ \Phi$ is the action of g on the output side. Stacking equivariant layers preserves equivariance, but any task that requires a prediction independent of the group action—classification, regression, detection—must eventually map equivariant features to *invariant* ones (constant on each orbit).

Table 1 summarises the trade-offs between exactness, completeness, and computational cost across existing invariant pooling layers. Norm pooling and gated nonlinearities are exact invariants but *incomplete*: they discard the phase of Fourier blocks, collapsing distinct orbits. Fourier-space pointwise nonlinearities preserve completeness but sacrifice exactness through

Table 1: Invariant nonlinearities for equivariant CNNs. "Exact"* means the invariant has no discretisation/aliasing error; "Complete" means it separates orbits; "Genericity assumption"† means completeness holds outside a Lebesgue-measure-zero algebraic subvariety. The selective bispectrum is the only entry that is simultaneously exact, complete, and linear-cost for finite groups; the $\mathrm{SO}(3)$-on-S^2 case is treated empirically (Sec. 6).

Nonlinearity	Exact?*	Complete?	Coefficients	Forward cost	Key limitation
Norm	Yes	No	$O(\|G\|)$	$O(\|G\|)$	Discards phase
Gate	Yes	No	$O(\|G\|)$	$O(\|G\|)$	Scalar bottleneck
Fourier pointwise	No	Yes‡	$O(\|G\|)$	$O(\|G\| \log \|G\|)$	Aliasing
Full bispectrum (finite G)	Yes	Yes	$O(\|G\|^2)$	$O(\|G\|^3)$	Quadratic cost
Full bispectrum ($\mathrm{SO}(3)/S^2$)	Yes	Yes	$\Theta(L^3)$	$O(L^4)$	Cubic cost
Selective bispectrum (finite G)	Yes	Yes	$O(\|G\|)$	$O(\|G\|^2)$	Genericity assumption†
Augmented selective $\mathrm{SO}(3)/S^2$ (ours)	Yes	Empirical	$\Theta(L^2)$	$O(L^3)$	Proof of concept only

‡Complete pre-aliasing; the pointwise nonlinearity introduces aliasing that may break completeness in practice.

aliasing [Weiler and Cesa, 2019]. The full bispectrum is exact and complete but costs $O(|G|^2)$ coefficients. The *selective* bispectrum (introduced below) preserves both exactness and completeness while using only $O(|G|)$ coefficients on generic signals.

2.2 The G-bispectrum and selective bispectra

Let G be a compact group with dual $\hat{G}$ and irreps $\{\rho_i\}_{i\in\hat{G}}$ of dimensions d_i. The *group Fourier transform* [Diaconis and Rockmore, 1990] of a signal $f : G \to \mathbb{C}$ is

$$\hat{f}(\rho_i) = \int_G f(g)\, \rho_i(g)\, d\mu(g) \quad \in \mathbb{C}^{d_i \times d_i}, \tag{1}$$

which specialises to $\sum_{g\in G} f(g)\, \rho_i(g)$ when G is finite (counting measure). The Clebsch–Gordan (CG) decomposition $\rho_i \otimes \rho_j \cong \bigoplus_{k\in\hat{G}} m_{ij}^k \rho_k$ expresses the tensor product of two irreps as a direct sum of irreps (with integer multiplicities $m_{ij}^k \geq 0$); the change-of-basis is realised by the *CG isometry* $C_{ij}^k : V_{\rho_k} \to V_{\rho_i} \otimes V_{\rho_j}$, projecting the tensor product onto its ρ_k-isotypic component. For each admissible triple (i, j, k) with $m_{ij}^k > 0$, the *bispectrum* is the scalar

$$\beta_{i,j}^k(f) := \left\langle\, (C_{ij}^k)^\dagger \left(\hat{f}(\rho_i) \otimes \hat{f}(\rho_j)\right),\, \hat{f}(\rho_k)\right\rangle. \tag{2}$$

Each $\beta_{i,j}^k$ is invariant under the action $f \mapsto f \circ g$. The full matrix/operator bispectrum is complete on compact groups under Kakarala-style hypotheses [Kakarala, 1992, 2012]; for finite G, the scalar-entry family $\{\beta_{i,j}^k\}_{(i,j,k)}$ remains complete on generic signals via the selective construction of Mataigne et al. [2024]. For $\mathrm{SO}(3)$ on S^2 we do not claim scalar completeness—the library's selective invariant is empirical; see Section 6. For finite G, $|\hat{G}| = O(|G|)$, so the full bispectrum has $O(|G|^2)$ scalar coefficients (indexed by $(i, j) \in \hat{G}^2$ with k ranging over the CG-allowed irreps). Equation (2) is stated for G acting on itself; for homogeneous spaces of the form G/H (e.g. $\mathrm{SO}(3)$ on S^2, or $\mathrm{SO}(2)$ on the disk), the same construction applies after expanding f in a basis adapted to G/H (spherical harmonics for S^2, Bessel modes on the disk). The explicit specialisation for S^2 is in Definition 3.

Selective bispectra. Mataigne et al. [2024] show that for finite groups, a BFS over the Kronecker product table of $\hat{G}$ selects $O(|G|)$ bispectral coefficients $\beta_{i,j}^k$ that remain complete; Myers and Miolane [2025] extend this to the disk. This reduction from quadratic to linear is the key to practical deployment: for example, the bispectrum of the octahedral group drops from 576 to 172 coefficients.

3 Related Work

Equivariant network libraries. `escnn` [Cesa et al., 2022] and `e3nn` [Geiger and Smidt, 2022] provide general-purpose equivariant convolution layers for 2D and 3D groups respectively, and `lie-nn` [Batatia et al., 2023] provides general Lie-group representation primitives (irreducible representations, Clebsch–Gordan decompositions, tensor products) on top of which custom equivariant

Table 2: Scope of the `bispectrum` library. `SO2onS1` is the continuous-n limit of `CnonCn` and shares its implementation.

Module	Group / Domain	Output mode	Reference
`CnonCn`	C_n on C_n	selective + full	[Sanborn et al., 2023]
`SO2onS1`	$\mathrm{SO}(2)$ on S^1	selective + full	[Mataigne et al., 2024]
`TorusDnTorus`	$\mathbb{T}^d$	selective + full	[Mataigne et al., 2024]
`DnonDn`	D_n on D_n	selective	[Mataigne et al., 2024]
`SO2onDisk`	$\mathrm{SO}(2)$ on disk	selective	[Myers and Miolane, 2025]
`SO3onS2`	$\mathrm{SO}(3)$ on S^2	selective + full	[Kakarala, 2012], this paper
`OctaonOcta`	chiral octahedral O	selective	[Mataigne et al., 2024]

layers can be built. These libraries implement equivariant *feature maps* but leave the choice of invariant pooling to the user (typically norm or gate). Our library is complementary: it provides the invariant map as a drop-in `nn.Module` that plugs into architectures built with any of these frameworks.

Bispectral approaches in neural networks. Sanborn et al. [2023] introduced bispectral neural networks (BNNs), using the full G-bispectrum as an invariant nonlinearity for C_n and D_n. Oreiller et al. [2022] applied bispectral CNNs to medical imaging with a fixed set of hand-selected triples. Both use the full (quadratic-cost) bispectrum and provide implementations tied to specific architectures. Our library generalizes to seven group/domain pairs, defaults to the $O(|G|)$ selective reduction [Mataigne et al., 2024], and exposes a framework-agnostic API. Chevalley et al. [2024] explore bispectral signatures for data characterisation, using bispectra as descriptors rather than learnable pooling primitives. Outside the equivariant-network setting, `PyBispectra` [Binns et al., 2025] computes bispectrum and bicoherence statistics of one-dimensional time-series signals for electrophysiology applications such as phase-amplitude coupling and time-delay estimation; the underlying 1D bispectrum is translation-invariant by construction, but the toolbox is not designed as a neural-network pooling layer.

Bispectra on homogeneous and non-homogeneous spaces. The selective G-bispectrum construction [Mataigne et al., 2024] reduces the cost of the G-bispectrum from $O(|G|^2)$ to $O(|G|)$ scalars on finite groups. Myers and Miolane [2025] extend the selective construction to the disk, where $\mathrm{SO}(2)$ acts on a non-homogeneous space. Kakarala [2012] treats the $\mathrm{SO}(3)$-on-$\mathrm{SO}(3)$ bispectrum and the closely related sphere case via Wigner-D irreps. The matrix (vector) bispectrum on S^2 is the homogeneous-space form on which our scalar contractions and CG power augmentation are built.

Invariant theory and scattering. The wavelet scattering transform [Mallat, 2012] provides translation-invariant representations via iterated modulus operations; Esteves et al. [2018] extend this to the sphere. Unlike the G-bispectrum, scattering representations are stable but generally incomplete—they discard inter-scale phase information. The G-bispectrum's completeness guarantee [Giannakis, 1989, Kakarala, 2012] is its distinguishing theoretical advantage.

4 The `bispectrum` Library

4.1 Design and implementation

Every G/domain pair is a `torch.nn.Module` named `{Group}on{Domain}` (e.g. `SO3onS2`); Table 2 lists all seven modules. Each exposes a uniform API: `forward(f)` returns the selective invariants, `fourier(f)` the group Fourier coefficients, and `invert(beta)` the reconstructed signal up to group-action indeterminacy (where available). Modules default to $O(|G|)$ selective coefficients (`selective=False` restores the full $O(|G|^2)$ set), precompute CG matrices / DFT kernels / Bessel roots as non-learnable buffers, and depend only on PyTorch [Paszke et al., 2019], NumPy [Harris et al., 2020], and `torch_harmonics` [Bonev et al., 2023] (for `SO3onS2`). Test suite line coverage is 96.8%; a minimal usage example is in Appendix F.

Table 3: Library profiling summary on a single NVIDIA H100 80 GB GPU. "Sel. coefs" = selective coefficient count; "Fwd" /"Inv" = median forward/inversion wall-clock (ms); "–" = not implemented. *`SO3onS2` at $\ell_{\max}$=16. The full bispectrum at matched group order is 8256/524,800 coefficients for $C_{128}/\mathbb{T}^2$ (vs. 128/1024 selective, a 64×/512× reduction), taking 9.17/0.71 ms in the forward pass.

Module	G	$\lvert G\rvert$	Sel. coefs	Fwd (ms)	Inv (ms)
`CnonCn`	C_{128}	128	128	0.15	8.56
`TorusOnTorus`	$C_{32}\times C_{32}$	1024	1024	0.08	68.02
`DnonDn`	D_{32}	64	245	2.52	42.33
`SO2onDisk`	$\mathrm{SO}(2)$	∞	411	1.94	17.69
`SO3onS2`*	$\mathrm{SO}(3)$	∞	430	26.54	–
`OctaonOcta`	O	24	172	0.70	–

Table 4: PCam at matched ∼100K params, 10% training data, mean ± std over 3 seeds; all equivariant models share a C_8-equivariant DenseNet backbone.

Model	Params	Test AUC	Test ACC
Standard (augmented)	102K	$.896 \pm .010$	$.826 \pm .012$
Gated	136K	$.941 \pm .009$	$.870 \pm .004$
Selective Bispectrum (`CnonCn`)	128K	$.941 \pm .004$	$\mathbf{.872 \pm .006}$
Fourier-ELU	110K	$\mathbf{.945 \pm .004}$	$.855 \pm .006$
NormReLU	110K	$.942 \pm .004$	$.858 \pm .007$

4.2 Profiling

We profile the library on a single NVIDIA H100 80 GB GPU (PyTorch 2.10, batch = 16 for the forward pass and 4 for inversion, median wall-clock via `torch.utils.benchmark`); results are in Table 3. Coefficient-count scaling and additional plots are in Appendix B (Figure 4).

The selective forward pass is sub-millisecond for finite abelian modules (C_n: 0.15 ms, $\mathbb{T}^2$: 0.08 ms), under 2 ms for $\mathrm{SO}(2)$/disk and under 1 ms for the octahedral group, making it negligible relative to backbone computation. $\mathrm{SO}(3)$ on S^2 at $\ell_{\max}$=16 takes 26.5 ms, dominated by the spherical harmonic transform (`torch_harmonics` [Bonev et al., 2023]). GPU throughput scales linearly with batch size, exceeding 10^7 samples/s for C_n and $\mathbb{T}^2$ (Appendix B). Six backends are tabulated; the seventh module, `SO2onS1`, subclasses `CnonCn` with $n = \ell_{\max}+1$ and has identical runtime to the corresponding `CnonCn` row at matched $\lvert G\rvert$.

5 Experiments

We validate the library on three tasks spanning discrete and continuous groups: 2D histopathology classification under C_8 symmetry (PCam, Section 5.1), 3D organ classification under octahedral symmetry (OrganMNIST3D, Section 5.2), and spherical digit classification under full $\mathrm{SO}(3)$ (Spherical MNIST, Section 5.3).

5.1 2D histopathology classification under cyclic symmetry

We evaluate five invariant pooling strategies on PatchCamelyon [Veeling et al., 2018] (PCam), 327K histopathology patches with binary metastasis labels. All equivariant models share a C_8-equivariant DenseNet [Huang et al., 2017] backbone following Veeling et al. [2018]; they differ only in the terminal invariant map: norm pooling, gated pooling, Fourier-ELU, and selective bispectrum via `CnonCn`(n=8). A non-equivariant standard CNN trained with data augmentation drawn uniformly from the symmetry group C_8 ("*augmented CNN*") serves as the non-equivariant baseline. We sweep growth rates to vary capacity (30K–1.6M params) and train on 10% of the PCam training set (an image-level subsample, not a group-orbit subsample, drawn before any group augmentation) with AdamW [Loshchilov and Hutter, 2019]. Full model descriptions are in Appendix C.

Table 4 reports multi-seed statistics at matched ∼100K params; the full Pareto sweep (AUC vs. parameter count, ranking, data efficiency) is Figure 7 in Appendix C.

Table 5: OrganMNIST3D results (mean $\pm$ std over 3 seeds). Rot ACC and σ_{rot}: mean and std of accuracy across 24 octahedral test-set rotations.

Model	Params	Test ACC	Test AUC	Rot ACC	σ_{rot}
Standard 3D CNN	16K	$.601 \pm .017$	$.951 \pm .002$	.598	.0115
O-Equiv + Norm Pool	375K	$.568 \pm .099$	$.940 \pm .021$	.568	.0000
O-Equiv + Max Pool	374K	$.730 \pm .033$	$.972 \pm .005$	.730	.0000
O-Equiv + `OctaonOcta` Bispectrum	463K	$.726 \pm .027$	$.972 \pm .004$	.726	.0005†

†Nonzero σ_{rot} due to floating-point precision in CG contractions, not a theoretical invariance failure.

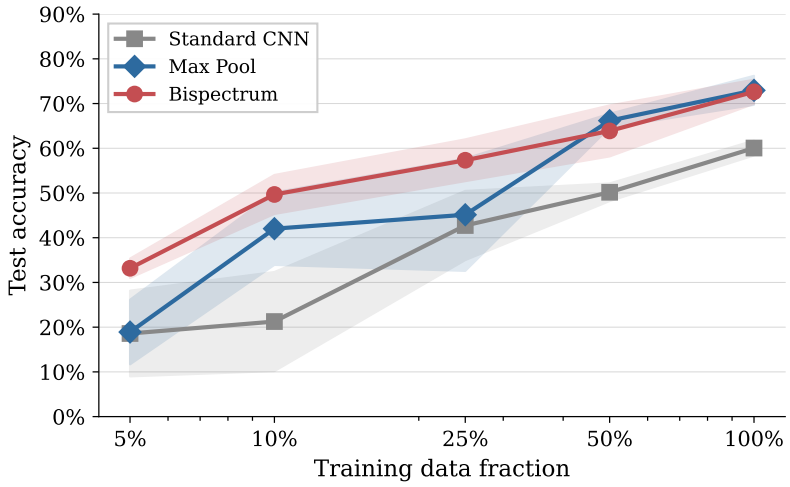

Figure 1: Test accuracy vs. training data fraction on OrganMNIST3D (`OctaonOcta` bispectrum, mean over 3 seeds; shaded bands: ± 1 std). The bispectrum provides a clear advantage in the low-data regime.

At matched capacity ($\sim$100K params, 10% data), all four equivariant pooling methods substantially outperform the standard augmented CNN (AUC 0.896). Fourier-ELU leads slightly (0.945), followed by NormReLU (0.942), bispectrum and gated (both 0.941). At 1% data, the bispectrum leads all methods (0.912), outperforming NormReLU (0.873) and Fourier-ELU (0.876). This data-efficiency advantage narrows as the training set grows, consistent with complete invariants providing a stronger inductive bias when data is scarce.

5.2 3D organ classification under octahedral symmetry

To evaluate the `OctaonOcta` module for the chiral octahedral group O ($|O| = 24$), we design a controlled 3D experiment on OrganMNIST3D [Yang et al., 2023] (1,742 abdominal CT volumes, 28^3 voxels, 11 classes). All equivariant variants share an O-equivariant 3D ResNet backbone with exact voxel-permutation kernels. The four models compared are: a translation-equivariant standard 3D CNN trained with octahedral data augmentation (the "*augmented CNN*" baseline, 16K params); the same O-equivariant ResNet with terminal norm pool (375K params); with terminal max pool (374K params); and with terminal selective bispectrum via `OctaonOcta` (463K, 172 invariant scalars). Models are trained with AdamW [Loshchilov and Hutter, 2019] for 100 epochs with three seeds; rotation robustness is measured over all 24 octahedral rotations of the test set. Published baselines and wider-channel experiments are in Appendix D.

Table 5 and Figure 1 report the key findings.

All equivariant models achieve $\sigma_{\text{rot}} \leq 0.0005$ while the augmented CNN drops $\sim$2.5 pp under rotation. The bispectrum (72.6%) matches max pool (73.0%) and substantially outperforms norm pool (56.8%) and Regular SE(3) convolutions [Kuipers and Bekkers, 2023] (69.8%). At reduced data (Figure 1), the bispectrum's advantage widens: at 5% data it reaches 33.2% vs. 18.9% for max pool. At higher capacity (6M params, Appendix D), max pool overtakes the bispectrum, suggesting the completeness advantage is specific to the data-limited regime.

Table 6: Spherical MNIST results (mean $\pm$ std over 3 seeds). Reference numbers are from Cohen et al. [2018]. σ_{rot}: standard deviation of accuracy under 10 random $\mathrm{SO}(3)$ rotations (NR-trained models).

Model	Params	NR/NR ↑	R/R ↑	NR/R ↑	σ_{rot} ↓
Cohen planar CNN	—	0.98	0.23	0.11	—
Cohen spherical CNN	—	0.96	0.95	0.94	—
Standard CNN	185K	0.991±0.000	0.781±0.003	0.138±0.003	0.044
Power Spec. + MLP	11K	0.780±0.002	0.780±0.002	0.777±0.001	3×10^{-4}
Power Spec. + MLP (matched)	166K	0.794±0.011	0.794±0.011	0.792±0.010	3×10^{-4}
`SO3onS2` **Bispectrum + MLP (ours)**	232K	**0.950±0.000**	**0.949±0.002**	**0.951±0.001**	$\mathbf{1\times10^{-4}}$

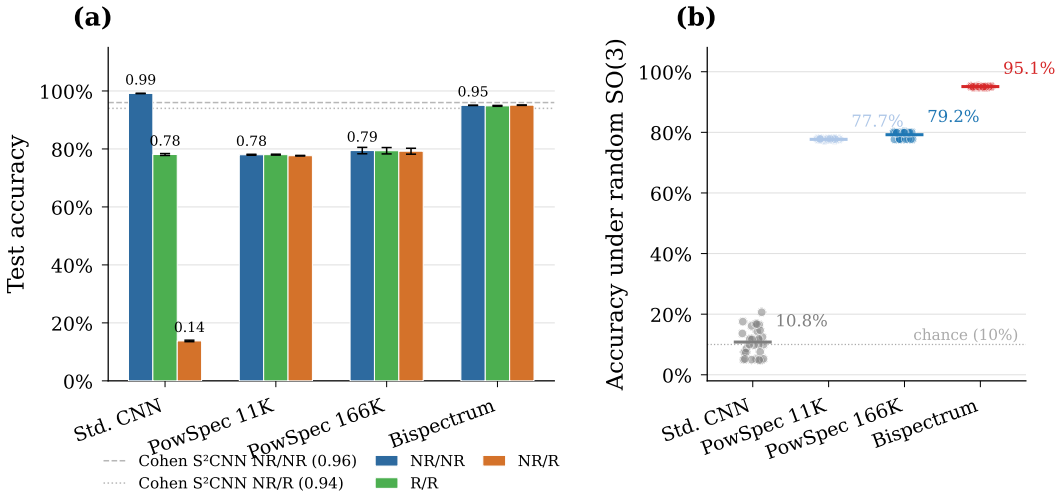

Figure 2: Spherical MNIST analysis (`SO3onS2` selective bispectrum). **(a)** Cohen et al. protocol: NR/NR, R/R, and NR/R accuracy. Dashed lines show Cohen's spherical CNN reference. **(b)** Accuracy under 10 random $\mathrm{SO}(3)$ rotations (NR-trained models, 30 points per model = 10 rotations $\times$ 3 seeds). Scaling the power spectrum MLP from 11K to 166K parameters yields only +1.4pp; the `SO3onS2` bispectrum at 232K reaches 95.0%.

5.3 Spherical MNIST: SO(3) invariance on the sphere

The preceding experiments validate bispectral pooling for finite groups (C_8 and the octahedral group O). To evaluate the library's continuous group support, we test `SO3onS2`—the augmented selective $\mathrm{SO}(3)$ invariant on the 2-sphere (Section 6)—on Spherical MNIST [Cohen et al., 2018], the standard benchmark for rotation invariance on S^2.

Setup. MNIST digits are stereographically projected onto a 64×128 equiangular grid; we compute Φ_{sel} at $\ell_{\max} = 15$ ($N = 384$ live real components, stored as 384 complex tensors by the library), apply sign-preserving $\mathrm{sign}(x)\log(1+|x|)$ to real and imaginary parts ($2N = 768$ real channels; one channel per bispectral entry is structurally zero by parity but we feed both since the log transform is per-channel injective and cheap), and feed a 3-layer MLP (232K params). Controls: a power-spectrum MLP at 11K params (equally invariant but *incomplete*, discarding inter-harmonic phase), a matched 166K-param version isolating capacity vs. completeness, and a non-invariant CNN directly on the equirectangular image at 185K params. Following Cohen et al. [2018], we evaluate under three protocols NR/NR, R/R, and the most challenging NR/R; each run uses three seeds, and rotation robustness σ_{rot} is measured over 10 random $\mathrm{SO}(3)$ rotations of the test set.

Table 6 and Figure 2 report the results.

The bispectrum achieves near-identical accuracy across all three protocols ($\sigma_{\text{rot}}\sim10^{-4}$), confirming true $\mathrm{SO}(3)$ invariance; the standard CNN collapses from 99.1% (NR/NR) to 13.8% (NR/R), showing that augmentation alone does not confer invariance. The power spectrum is equally invariant

but reaches only 78%—a 17pp deficit that persists after scaling the MLP to 166K params ($+1.4$pp), consistent with the power spectrum's incompleteness (rather than MLP capacity) being the bottleneck. Cohen's spherical CNN [Cohen et al., 2018] reaches $96\%/95\%/94\%$ with learned equivariant convolutions; our single-pass bispectrum ($95.0\%/94.9\%/95.1\%$) is 1–2pp lower but has a tighter invariance gap (0.1pp vs. 2.0pp).

6 Augmented selective $\mathrm{SO}(3)$ invariant on S^2

6.1 Setup and construction

A real signal $f : S^2 \to \mathbb{R}$ band-limited at degree L has spherical harmonic coefficients $\mathbf{F}_\ell \in \mathbb{C}^{2\ell+1}$, $\ell = 0, \ldots, L$, transforming as $\mathbf{F}_\ell \mapsto D^\ell(g)\mathbf{F}_\ell$ under $g \in \mathrm{SO}(3)$; the reality condition leaves $(L{+}1)^2$ real degrees of freedom (Appendix A.1). The scalar bispectral coefficient at a CG-admissible triple (ℓ_1, ℓ_2, ℓ) is

$$\beta_{\ell_1,\ell_2,\ell} = \sum_{\substack{m_1,m_2,m \\ m_1+m_2=m}} C^{\ell,\,m}_{\ell_1,\,m_1;\,\ell_2,\,m_2}\, a^{m_1}_{\ell_1}\, a^{m_2}_{\ell_2}\, \overline{a^{m}_{\ell}}, \tag{3}$$

the specialisation of (2) to $\mathrm{SO}(3)/S^2$. The full set has $\Theta(L^3)$ scalars; the orbit space $\mathbb{R}^{(L+1)^2}/\mathrm{SO}(3)$ has dimension $(L{+}1)^2 - 3$, so any smooth complete invariant requires $\Omega(L^2)$ components (Proposition 7), which the Mataigne chain $\beta_{1,\ell,\ell+1}$ ($\sim 3L$ scalars) cannot achieve.

We define $\Phi_{\mathrm{sel}}(f)$ as the output of the augmented selective index construction of Algorithm 2 (Appendix A.3). For each degree ℓ it collects (i) a size-$(2\ell{+}1)$ linear bootstrap block $\mathcal{T}_\ell$ of scalar bispectrum entries, (ii) mandatory even self-couplings $\beta_{\ell,\ell,\ell'}$, and (iii) a greedily chosen set of CG-power scalars $P_{\ell_1,\ell_2,\ell} := \|(\mathbf{F}_{\ell_1} \otimes \mathbf{F}_{\ell_2})|_\ell\|^2$ (the bootstrap block alone is rank-deficient on real signals, see Remark 18). Because Φ_{sel} mixes bispectrum entries with CG-power scalars (a degree-4 invariant), we call it an *augmented selective invariant* rather than a bispectrum; the phrase "selective bispectrum" refers strictly to the bispectral component $\mathcal{S}_\beta$. For degrees $\ell \le L_{\mathrm{seed}} := \min(L, 4)$ we additionally include the full seed $\mathcal{S}_{\mathrm{seed}}$. The total size is $|\Phi_{\mathrm{sel}}| = \Theta(L^2)$ (Proposition 11); Algorithm 1 summarises the bispectral-triple selection, and the full index set—including self-couplings and CG power—is laid out in Algorithm 2. Concretely, at $L = 15$ (resp. $L = 16$) the library outputs $|\Phi_{\mathrm{sel}}| = 384$ (resp. 430) real scalars; the full bispectrum at $L = 15$ has 2,056 admissible triples (cf. Table 8).

Algorithm 1 Selected bispectral triples $\mathcal{S}_\beta$ for the augmented $\mathrm{SO}(3)$ invariant (BFS form; mirrors Mataigne et al. [2024], Alg. 1). The full index set — bispectral triples, mandatory self-couplings, and CG-power scalars — is Algorithm 2 in Appendix A.3.

```
Input: band-limit L, seed cutoff L_seed = min(L, 4)
L_β ← all admissible triples (ℓ1, ℓ2, ℓ) with ℓ ≤ L_seed {seed block S_seed}
for ℓ = L_seed+1, ..., L do
    append the bootstrap block T_ℓ to L_β {2ℓ+1 entries linear in F_ℓ; tabulated in Appendix A.3}
end for
return {β_{ℓ1,ℓ2,ℓ}(f) : (ℓ1, ℓ2, ℓ) ∈ L_β}
```

6.2 Empirical proof of concept: bispectrum reconstruction

To test whether Φ_{sel} retains enough information to recover signals up to rotation, we run a direct inversion experiment on Spherical MNIST digits band-limited at $L = 12$.[1] For each source signal $f : S^2 \to \mathbb{R}$ we apply a uniform random rotation $R \in \mathrm{SO}(3)$ to form the target $R \cdot f$, compute $\Phi_{\mathrm{sel}}(R \cdot f)$, and recover $\hat{f}$ by gradient descent from a random Gaussian initialisation, minimising $\|\Phi_{\mathrm{sel}}(\hat{f}) - \Phi_{\mathrm{sel}}(R \cdot f)\|^2 / \|\Phi_{\mathrm{sel}}(R \cdot f)\|^2$ (Adam, multi-restart, cosine-annealed LR). The orbit ambiguity is then resolved by Procrustes-style optimisation on a quaternion parameterisation: we fit $\hat{R} \in \mathrm{SO}(3)$ minimising $\|\hat{R} \cdot \hat{f} - R \cdot f\|^2$.

[1] We use $L{=}12$ here rather than the classifier's $L{=}15$ to keep the reconstruction residual safely below the SHT discretisation floor at the $64{\times}128$ grid; protocol details are in Appendix E.5.

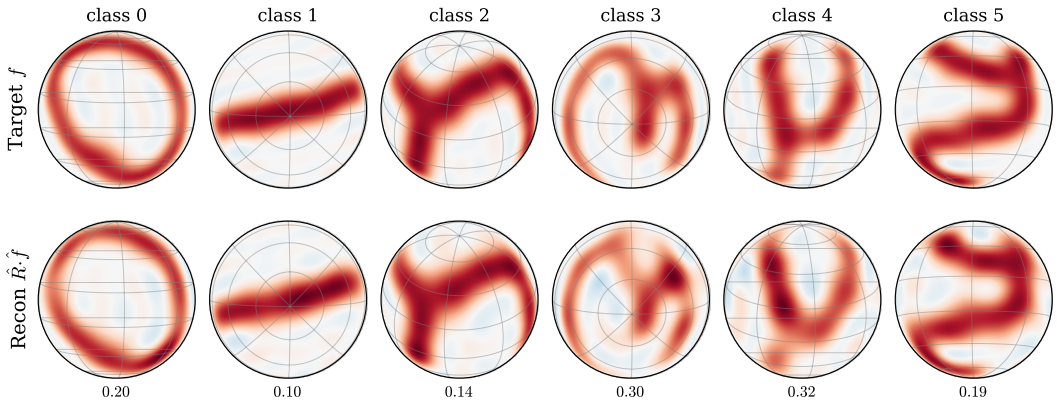

Figure 3: Φ_{sel} reconstruction (proof of concept). Top: target f (digit classes 0–5). Bottom: $\hat{R}\cdot\hat{f}$, recovered from $\Phi_{\text{sel}}(f)$ by gradient descent and Procrustes-aligned. Numbers: relative image residual $\|\hat{R}\cdot\hat{f}-f\|/\|f\|$, median $\approx$0.20 across the six classes (raw pre-alignment residual $\approx$1.3).

Takeaways. Bispectrum residuals reach $\|\Delta\Phi\|/\|\Phi\| \sim 10^{-4}$–$10^{-3}$ (below the SHT-discretisation invariance noise floor of $\sim 8\times 10^{-3}$ at this grid); the aligned image residuals $\|\hat{R}\cdot\hat{f}-f\|/\|f\| \sim 0.1$–$0.3$ are consistent with local-optimisation error or conditioning effects, and we did not observe a systematic missing-information failure mode, but the experiment does not rule out incompleteness or near-collisions in feature space. Visually, $\hat{R}\cdot\hat{f}$ aligns with f up to rotation across all six digit classes shown (Figure 3; the same behaviour holds across our 8-digit sweep, see Appendix E.5). We read this as proof-of-concept evidence that Φ_{sel} preserves substantial orbit information on band-limited signals; a formal completeness proof remains open.

7 Discussion and Conclusion

We have presented `bispectrum`, a PyTorch library that makes selective G-bispectral invariants practical, covering seven group/domain pairs through a uniform `nn.Module` interface, and we contribute a $\Theta(L^2)$ selective $\mathrm{SO}(3)$ invariant on S^2 (Section 6). The selective forward pass adds negligible overhead ($<$1 ms for finite-group modules) and delivers numerically exact rotation invariance with a consistent data-efficiency advantage over incomplete alternatives in the low-data / moderate-capacity regime.

When not to use the bispectrum. At high model capacity (Sec. 5), incomplete alternatives like max pooling can match or surpass bispectral pooling, and each module is invariant to a *specific* group only—data augmentation or flexible-symmetry equivariant networks fit better when the relevant symmetry is multi-group or unknown.

Limitations and future work. Inversion from Φ_{sel} is not shipped as a library API, and a formal completeness proof, a characterisation of the degenerate set, and numerical-conditioning analysis at large band-limits remain open. We do not compare against tensor-product architectures (e3nn, MACE) or wavelet scattering. Natural next steps are $\mathrm{SO}(3)$ inversion and cryo-EM / molecular applications.

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A Selective $\mathrm{SO}(3)$ Bispectrum – Structural Background

This appendix gives the structural background for the construction of Section 6: the S^2 band-limited setup and notation (Section A.1); the orbit-space dimension lower bound $(L{+}1)^2 - 3$ that motivates the $\Theta(L^2)$ target (Section A.2); the explicit selective index set together with the implementation-grade BFS variant (Algorithm 2) and per-degree counting result (Section A.3); and the parity structure of the scalar bispectrum that dictates the exclusion of certain odd-parity triples in $\mathcal{T}_\ell$ (Section A.4). A short comparison with prior bispectral constructions (Section A.5) and a pixel-space complexity table (Section A.6) close the appendix. We do not claim a closed-form completeness proof in this revision; the empirical reconstruction of Figure 3 is the proof-of-concept evidence the construction is supported by.

Per-degree entry allocation: the full seed $\mathcal{S}_{\mathrm{seed}}$ handles $\ell \le L_{\mathrm{seed}} = 4$ in constant size; for $\ell \ge 5$, Algorithm 2 allocates $2\ell{+}1$ bispectral rows plus $O(\ell)$ self-coupling and CG-power entries, giving $|\Phi_{\mathrm{sel}}| = \Theta(L^2)$ total (matching the orbit-space lower bound $(L{+}1)^2 - 3$ of Proposition 7; exact per-L counts in Table 8).

A.1 Setup and notation

Definition 1 (Band-limited signal on S^2)**.** A signal $f : S^2 \to \mathbb{R}$ is *band-limited* at degree L if its spherical harmonic expansion

$$f(\theta, \varphi) = \sum_{\ell=0}^{L} \sum_{m=-\ell}^{\ell} a_\ell^m \, Y_\ell^m(\theta, \varphi) \tag{4}$$

has $a_\ell^m = 0$ for all $\ell > L$. We write $\mathbf{F}_\ell = (a_\ell^{-\ell}, \ldots, a_\ell^{\ell}) \in \mathbb{C}^{2\ell+1}$ for the coefficient vector at degree ℓ.

For real f, the conjugation symmetry $a_\ell^{-m} = (-1)^m \overline{a_\ell^m}$ holds, so $\mathbf{F}_\ell$ has $2\ell + 1$ real degrees of freedom. The total signal dimension is $\dim = \sum_{\ell=0}^{L}(2\ell + 1) = (L + 1)^2$.

Definition 2 (Rotation action)**.** Under $g \in \mathrm{SO}(3)$, the spherical harmonic coefficients transform as $\mathbf{F}_\ell \mapsto D^\ell(g)\, \mathbf{F}_\ell$, where D^ℓ is the $(2\ell + 1)$-dimensional Wigner D-matrix.

In the irrep notation of Mataigne et al. [2024], $\{\rho_\ell\}_{\ell \ge 0}$ are the (unitary) irreducible representations of $\mathrm{SO}(3)$ with V_{ρ_ℓ} the $(2\ell{+}1)$-dimensional space of degree-ℓ spherical harmonics and $\rho_\ell(g) = D^\ell(g)$; the Fourier coefficient of f at ρ_ℓ is $\mathcal{F}(f)_{\rho_\ell} = \mathbf{F}_\ell$. We use $\mathbf{F}_\ell$ as the working symbol throughout.

Definition 3 (Bispectrum on S^2)**.** For each triple (ℓ_1, ℓ_2, ℓ) satisfying the triangle inequality $|\ell_1 - \ell_2| \le \ell \le \ell_1 + \ell_2$, the *bispectral coefficient* is

$$\beta_{\ell_1,\ell_2,\ell} = \sum_{\substack{m_1,\, m_2,\, m \\ m_1+m_2=m}} C^{\ell,\, m}_{\ell_1,\, m_1;\, \ell_2,\, m_2} \; a_{\ell_1}^{m_1} \, a_{\ell_2}^{m_2} \, \overline{a_\ell^m}, \tag{5}$$

where $C^{\ell,\, m}_{\ell_1,\, m_1;\, \ell_2,\, m_2}$ denotes the Clebsch–Gordan coefficient $\langle \ell_1, m_1;\, \ell_2, m_2 \mid \ell, m \rangle$. Each $\beta_{\ell_1,\ell_2,\ell}$ is a scalar; for real f it is real when $\ell_1{+}\ell_2{+}\ell$ is even and purely imaginary when $\ell_1{+}\ell_2{+}\ell$ is odd (Lemma 12).

Definition 3 is the specialization of the general G-bispectrum (eq. (2)) to the case $G = \mathrm{SO}(3)$ acting on S^2, where the irreps are Wigner-D matrices and the Fourier coefficients are spherical harmonic coefficients a_ℓ^m.

Definition 4 (CG power spectrum, optional diagnostic)**.** For a triple (ℓ_1, ℓ_2, ℓ) satisfying the triangle inequality, write $(\mathbf{F}_{\ell_1} \otimes \mathbf{F}_{\ell_2})\big|_\ell$ for the projection of the Kronecker product $\mathbf{F}_{\ell_1} \otimes \mathbf{F}_{\ell_2}$ onto the D^ℓ-isotypic component, realised by contracting with the CG coefficients $C^{\ell,\, m}_{\ell_1,\, m_1;\, \ell_2,\, m_2}$. The *CG power spectrum entry* is the squared norm of this projection,

$$P_{\ell_1,\ell_2,\ell} := \|(\mathbf{F}_{\ell_1} \otimes \mathbf{F}_{\ell_2})|_\ell\|^2 = \sum_{m=-\ell}^{\ell} \Big| \sum_{\substack{m_1, m_2 \\ m_1+m_2=m}} C^{\ell,\, m}_{\ell_1,\, m_1;\, \ell_2,\, m_2} \; a_{\ell_1}^{m_1} \, a_{\ell_2}^{m_2} \Big|^2. \tag{6}$$

This is a degree-4, real-valued, $\mathrm{SO}(3)$-invariant polynomial in the signal coefficients. The library appends a $O(L^2)$ family of CG power scalars to the selective bispectrum unconditionally for numerical robustness on the real-signal slice (Remark 18); they leave the asymptotic output size unchanged.

Proposition 5 (Size of the full bispectrum)**.** *The number of admissible triples* (ℓ_1, ℓ_2, ℓ) *with* $0 \leq \ell_1, \ell_2, \ell \leq L$ *satisfying the triangle inequality* $|\ell_1 - \ell_2| \leq \ell \leq \ell_1 + \ell_2$ *is* $\Theta(L^3)$.

Proof. For fixed ℓ_1 and ℓ_2, the triangle inequality admits $\min(\ell_1+\ell_2, L) - |\ell_1 - \ell_2| + 1$ valid values of ℓ. Summing over $(\ell_1, \ell_2) \in [0, L]^2$ gives $N_{\text{full}} = \Theta(L^3)$. □

Remark 6 (Invariance). Each $\beta_{\ell_1,\ell_2,\ell}$ is $\mathrm{SO}(3)$-invariant: the Wigner matrices satisfy $D^{\ell_1} \otimes D^{\ell_2} = \bigoplus_\ell D^\ell$ under the CG decomposition, so the triple product contracted with CG coefficients is unchanged by $\mathbf{F}_\ell \mapsto D^\ell(g)\mathbf{F}_\ell$.

A.2 Why $\mathcal{O}(L)$ is impossible

Proposition 7 (Dimensional lower bound)**.** *Any* smooth or polynomial *complete* $\mathrm{SO}(3)$*-invariant for band-limited signals on* S^2 *at band-limit* L *must have codomain of dimension at least* $(L+1)^2-3$.

Proof. Let $\Phi : \mathbb{R}^{(L+1)^2} \to \mathbb{R}^k$ be a smooth (or polynomial) $\mathrm{SO}(3)$-invariant map that separates orbits on a Zariski-open dense subset $U \subset \mathbb{R}^{(L+1)^2}$. Smoothness implies Φ has a well-defined generic rank $r := \max_{x \in U} \operatorname{rank} d\Phi_x$; on the open dense subset where this rank is attained, Φ is locally a submersion onto an r-dimensional submanifold, and orbit-separation forces $r \geq \dim(\mathbb{R}^{(L+1)^2}/\mathrm{SO}(3)) = (L+1)^2 - \dim \mathrm{SO}(3) = (L+1)^2 - 3$ (generic-rank theorem for $\mathrm{SO}(3)$ acting on V_L; cf. Tarski–Seidenberg in the polynomial case). Hence $k \geq r \geq (L+1)^2 - 3 = \Omega(L^2)$. □

Remark 8. The chain $\beta_{1,\ell,\ell+1}$ for $\ell = 0, \ldots, L-1$ provides only $\sim 3L$ scalar constraints (since each chain entry couples $\mathbf{F}_1 \in \mathbb{C}^3$ with one known $\mathbf{F}_\ell$). This cannot recover $(L+1)^2 \approx L^2$ unknowns.

Remark 9 (Smoothness assumption). The smoothness/polynomial qualifier is essential: pathological non-measurable invariants (e.g. a fixed orbit-section indexed by the axiom of choice) can in principle have codomain dimension 1 but are not implementable. All practical bispectral or spectral invariants are polynomial, so the proposition applies.

A.3 The selective index set

For each target degree ℓ, the library's selective invariant allocates $2\ell+1$ linear bispectral entries, plus quadratic/cubic self-couplings and CG-power scalars for numerical robustness on the real-signal slice. Combined with a constant-size full-bispectrum seed at degrees $\leq L_{\text{seed}} = \min(L, 4)$, this gives an $O(L^2)$-sized invariant matching the orbit-space lower bound of Proposition 7.

Definition 10 (Augmented selective $\mathrm{SO}(3)$-on-S^2 invariant)**.** Set $L_{\text{seed}} := \min(L, 4)$. Let $\mathcal{S}_{\text{seed}}$ denote the bispectral triples at degrees $\ell \leq L_{\text{seed}}$ returned by Algorithm 2's seed phase: the pre-prune set collects all admissible triples $\{(\ell_1, \ell_2, \ell) : 0 \leq \ell_1 \leq \ell_2 \leq L_{\text{seed}}, |\ell_1 - \ell_2| \leq \ell \leq \min(\ell_1+\ell_2, L_{\text{seed}})\}$ (42 triples at $L_{\text{seed}} = 4$); Algorithm 2 then parity-prunes entries that vanish identically on real signals (Lemma 12), deduplicates, and caps each degree to its per-degree budget. The surviving bispectral count at $L = 4$ is 24 (see Table 8, column "Bispectral"), and $|\mathcal{S}_{\text{seed}}|$ is constant in L for $L \geq 4$. For each intermediate target $4 \leq \ell \leq 7$ Algorithm 2 uses the small explicit linear block $\mathcal{T}_\ell^{\text{small}}$ (Table 7); for each high-degree target $\ell \geq 8$ the *bootstrap block* $\mathcal{T}_\ell$ is

$$\begin{aligned} \mathcal{T}_\ell := \{&(a, \ell, \ell - a) : 1 \leq a \leq \ell-1\} \\ &\cup \big\{(a, \ell, \ell - a + 1) : 2 \leq a \leq \ell-1,\ a \neq (\ell+1)/2\big\} \\ &\cup \{(a, \ell - a, \ell) : 1 \leq a \leq 4\} \ \cup\ \mathcal{Z}_\ell, \end{aligned} \tag{7}$$

where $\mathcal{Z}_\ell = \emptyset$ for even ℓ and $\mathcal{Z}_\ell = \{(2, \ell-1, \ell)\}$ for odd $\ell \geq 9$. The exclusion $a \neq (\ell+1)/2$ for odd ℓ removes the self-conjugate triple $(r, 2r-1, r)$ at $\ell = 2r-1$, which has a repeated index and vanishes identically on real signals (Proposition 14); the compensating row $(2, \ell-1, \ell) \in \mathcal{Z}_\ell$ has all-distinct indices and does not overlap the chain family (which only covers pairs $(a, \ell-a)$ with $a \leq 4$). Size bookkeeping: $(\ell-1) + (\ell-2) + 4 = 2\ell+1$ for even ℓ, and $(\ell-1) + (\ell-3) + 4 + 1 = 2\ell+1$

for odd $\ell \geq 9$; the first two families differ in the last index ($\ell - a$ vs. $\ell - a + 1$), and $\mathcal{Z}_\ell$ at $a = 2$ has second coordinate $\ell - 1$ rather than $\ell - 2$, so all three subfamilies are disjoint.

The *augmented selective* $\mathrm{SO}(3)$*-on-*S^2 *invariant* Φ_{sel} is the output of Algorithm 2: bispectral triples $\mathcal{S}_\beta = \mathcal{S}_{\mathrm{seed}} \cup \bigcup_{\ell=5}^{L} \mathcal{T}_\ell$ (with $\mathcal{T}_\ell = \mathcal{T}_\ell^{\mathrm{small}}$ for $5 \leq \ell \leq 7$), mandatory even self-couplings $\beta_{\ell,\ell,\ell'}$ (ℓ' even, $2 \leq \ell' \leq \ell$), and CG-power scalars $P_{\ell_1,\ell_2,\ell} = \|(\mathbf{F}_{\ell_1} \otimes \mathbf{F}_{\ell_2})|_\ell\|^2$ (Definition 4) chosen greedily for per-degree Jacobian rank. Mathematically, $\Phi_{\mathrm{sel}} : V_L \to \mathbb{R}^N$ has N live real scalar components: by Lemma 12, each bispectral entry contributes one real component on the real-signal slice (real for even parity, purely imaginary for odd), and CG-power entries are real-valued degree-4 invariants. The PyTorch implementation stores each scalar in a complex tensor, so downstream layers receive $2N$ real channels after splitting real and imaginary parts; for bispectral entries exactly one of the two channels is structurally zero by parity, and for CG-power entries the imaginary channel is structurally zero.

Table 7: Explicit low-degree bootstrap blocks $\mathcal{T}_\ell^{\mathrm{small}}$ for $4 \leq \ell \leq 7$, copied from `bispectrum.so3_on_s2._small_linear_bootstrap_block`. Each triple (ℓ_1, ℓ_2, ℓ_3) denotes $\beta_{\ell_1,\ell_2,\ell_3}$; all are admissible and have ℓ in at least one slot. Entry counts $|\mathcal{T}_4^{\mathrm{small}}| = 10$, $|\mathcal{T}_5^{\mathrm{small}}| = 11$, $|\mathcal{T}_6^{\mathrm{small}}| = 13$, $|\mathcal{T}_7^{\mathrm{small}}| = 15$ match the per-degree budgets. $\mathcal{T}_4^{\mathrm{small}}$ is implementation-only: Algorithm 2 uses it as a ranked candidate list at $\ell = 4$, deduplicated against the seed $\mathcal{S}_{\mathrm{seed}}$ and capped to a budget of 10 entries; $\mathcal{T}_5^{\mathrm{small}}, \mathcal{T}_6^{\mathrm{small}}, \mathcal{T}_7^{\mathrm{small}}$ are the formal bootstrap blocks at $\ell \in \{5, 6, 7\}$.

ℓ	$\mathcal{T}_\ell^{\mathrm{small}}$ (each entry a triple (ℓ_1, ℓ_2, ℓ_3))
4	$(1,3,4)$, $(2,2,4)$, $(2,3,4)$, $(3,3,4)$, $(1,4,3)$, $(2,4,2)$, $(3,4,1)$, $(2,4,3)$, $(3,4,2)$, $(3,4,3)$
5	$(1,4,5)$, $(2,3,5)$, $(2,4,5)$, $(3,4,5)$, $(1,5,4)$, $(2,5,3)$, $(3,5,2)$, $(4,5,1)$, $(2,5,4)$, $(3,5,4)$, $(4,5,4)$
6	$(1,5,6)$, $(2,4,6)$, $(3,3,6)$, $(3,4,6)$, $(1,6,5)$, $(2,6,4)$, $(3,6,3)$, $(4,6,2)$, $(5,6,1)$, $(2,6,5)$, $(3,6,5)$, $(4,6,5)$, $(5,6,5)$
7	$(1,6,7)$, $(2,5,7)$, $(3,4,7)$, $(4,5,7)$, $(1,7,6)$, $(2,7,5)$, $(3,7,4)$, $(4,7,3)$, $(5,7,2)$, $(6,7,1)$, $(2,7,6)$, $(3,7,6)$, $(4,7,6)$, $(5,7,6)$, $(6,7,6)$

Table 8: Coefficient counts of Φ_{sel} at representative band-limits. "Bispectral" counts the triples in $\mathcal{S}_\beta$ (including mandatory even self-couplings); "CG-power" counts entries in $\mathcal{S}_P$; "Total N" is $|\Phi_{\mathrm{sel}}|$ (live real components) and matches the library's `output_size`. "Stored channels" counts the complex-tensor storage ($2N$) seen by the SMNIST classifier; one channel per bispectral entry is structurally zero by parity, and the imaginary channel of each CG-power entry is structurally zero.

L	Bispectral	CG-power	Total $N = \lvert\Phi_{\mathrm{sel}}\rvert$	Stored channels ($2N$)
4	24	10	34	68
5	37	17	54	108
15	307	77	384	768
16	348	82	430	860

Proposition 11 (Output size)**.** *For every $L \geq 4$, Algorithm 2 produces a $\Theta(L^2)$-sized invariant: the per-degree bootstrap block contributes at most $2\ell+1$ bispectral triples (Eq.* (7) *for $\ell \geq 8$ plus Table 7 for $5 \leq \ell \leq 7$), and the self-coupling and CG-power augmentations add $O(\ell)$ per degree, giving a total of $\Theta(L^2)$ real scalars. Per-L counts are listed in Table 8; both panels match the orbit-space lower bound $(L+1)^2 - 3$ of Proposition 7 up to a constant factor.*

A.4 Parity structure of the bispectrum

The selective construction of Section 6 exploits two structural facts about the scalar bispectrum $\beta_{\ell_1,\ell_2,\ell}$: a parity rule that fixes whether each entry is real or purely imaginary on real signals (Lemma 12) and the resulting identical vanishing of certain odd-parity entries with a repeated index (Proposition 14). The latter dictates the explicit exclusion in $\mathcal{T}_\ell$ at odd target degrees, and the former is the source of the rank deficit that motivates the CG-power augmentation of Algorithm 2 (Remark 18).

Lemma 12 (Parity of the bispectrum)**.** *For any signal (real or complex):*

$$\beta_{\ell_1,\ell_2,\ell}(T_R \cdot f) = (-1)^{\ell_1+\ell_2+\ell}\, \beta_{\ell_1,\ell_2,\ell}(f). \tag{8}$$

For real *signals, there is an independent constraint:*

$$\overline{\beta_{\ell_1,\ell_2,\ell}(f)} = (-1)^{\ell_1+\ell_2+\ell}\, \beta_{\ell_1,\ell_2,\ell}(f). \tag{9}$$

Together:

(a) If $\ell_1 + \ell_2 + \ell$ is even: β is real-valued (by (9)) and T_R-invariant (by (8)).
(b) If $\ell_1 + \ell_2 + \ell$ is odd: β is purely imaginary (by (9)) and $\beta(T_R \cdot f) = -\beta(f)$, so $\operatorname{Im}\beta$ flips sign under T_R.

Proof. Under T_R, each $a_\ell^m \mapsto (-1)^m a_\ell^{-m}$. Substituting into $\beta = \sum_m (\sum_{m_1+m_2=m} C\, a^{m_1} a^{m_2}) \overline{a^m}$, the three sign factors $(-1)^{m_1+m_2+m} = (-1)^{2m} = 1$ cancel. Re-indexing $m \mapsto -m$ and applying the CG symmetry $C^{\ell,\,-m}_{\ell_1,\,-m_1;\,\ell_2,\,-m_2} = (-1)^{\ell_1+\ell_2-\ell}\, C^{\ell,\,m}_{\ell_1,\,m_1;\,\ell_2,\,m_2}$ yields (8) (with $(-1)^{\ell_1+\ell_2-\ell} = (-1)^{\ell_1+\ell_2+\ell}$).

For real signals, $\overline{a_\ell^m} = (-1)^m a_\ell^{-m}$. Taking the complex conjugate of β and re-indexing similarly yields (9). □

Lemma 13 (CG power is T_R-invariant)**.** *For all triples (ℓ_1, ℓ_2, ℓ): $P_{\ell_1,\ell_2,\ell}(T_R \cdot f) = P_{\ell_1,\ell_2,\ell}(f)$.*

Proof. Under T_R, the CG-projected component transforms as $(\mathbf{F}_{\ell_1} \otimes \mathbf{F}_{\ell_2})|_\ell^M \mapsto (-1)^{\ell_1+\ell_2+\ell+M}\, (\mathbf{F}_{\ell_1} \otimes \mathbf{F}_{\ell_2})|_\ell^{-M}$. Taking the squared norm and re-indexing the sum over M leaves P unchanged. □

Proposition 14 (Vanishing of odd-parity entries with a repeated index)**.** *Let $\beta_{\ell_1,\ell_2,\ell}$ be the scalar bispectrum entry of a real signal f (so $\ell_1 + \ell_2 + \ell$ odd). If the index multiset $\{\ell_1, \ell_2, \ell\}$ contains a repeated value, then $\beta_{\ell_1,\ell_2,\ell}(f) \equiv 0$ as a polynomial in the SH coefficients of f.*

Proof. Define the symmetric scalar 3j contraction

$$T_{\ell_1,\ell_2,\ell_3}(f) := \sum_{m_1,m_2,m_3} \begin{pmatrix} \ell_1 & \ell_2 & \ell_3 \\ m_1 & m_2 & m_3 \end{pmatrix} a_{\ell_1}^{m_1}\, a_{\ell_2}^{m_2}\, a_{\ell_3}^{m_3}.$$

Converting the Wigner 3j into a Clebsch-Gordan coefficient via $\left(\begin{smallmatrix} j_1 & j_2 & j_3 \\ m_1 & m_2 & m_3 \end{smallmatrix}\right) = \frac{(-1)^{j_1-j_2-m_3}}{\sqrt{2j_3+1}}\, C^{j_3,\,-m_3}_{j_1,\,m_1;\,j_2,\,m_2}$ and applying the reality identity $a_{\ell_3}^{m_3} = (-1)^{m_3}\, \overline{a_{\ell_3}^{-m_3}}$ shows

$$\beta_{\ell_1,\ell_2,\ell}(f) = (-1)^{\ell_1+\ell_2} \sqrt{2\ell+1}\, T_{\ell_1,\ell_2,\ell}(f),$$

so β and T have the same vanishing locus. The 3j symbol is invariant under even column permutations and picks up $(-1)^{\ell_1+\ell_2+\ell_3}$ under odd ones. If two columns i, j satisfy $\ell_i = \ell_j$, the inserted vectors a_{ℓ_i}, a_{ℓ_j} are literally the same coefficient vector of f, so swapping those two columns leaves the contracted sum T unchanged while multiplying the 3j prefactor by $(-1)^{\ell_1+\ell_2+\ell_3}$. For odd parity this forces $T = -T$, hence $T \equiv 0$ and $\beta \equiv 0$. □

Remark 15 (Consequences for $\mathcal{T}_\ell$). Proposition 14 applies in particular to the family $(r, 2r-1, r)$: it has parity $4r-1$ (odd) and a repeated index, so $\beta_{r,2r-1,r} \equiv 0$ on real signals. Numerical evaluation at the seed-42 real witness confirms this: for $r = 2, \ldots, 7$ the entry is zero to double-precision rounding ($|\beta| \lesssim 10^{-14}$). Any formal definition of $\mathcal{T}_\ell$ must therefore exclude these triples (see (7)); we substitute a chain row with all-distinct indices in their place.

Corollary 16 (First nonzero odd-parity entry)**.** *The first bispectral entry with odd parity and generically nonzero imaginary part occurs at degree $\ell = 4$: $\beta_{2,3,4}$, with $\ell_1 + \ell_2 + \ell = 9$ (odd) and ℓ_1, ℓ_2, ℓ all distinct. At the deterministic witness (seed 42), $\operatorname{Im}(\beta_{2,3,4}) \approx 3.26 \neq 0$, and $\operatorname{Im}(\beta_{2,3,4})(T_R \cdot f) \approx -3.26$. Similarly, $\beta_{2,4,3}$ and $\beta_{3,4,2}$ are nonzero odd-parity entries at degree 4.*

Remark 17 (No parity-breaking at $\ell \leq 3$). For degrees ≤ 3, every odd-parity triple (ℓ_1, ℓ_2, ℓ) has at least two indices equal (since $\max(\ell_1, \ell_2, \ell) \leq 3$ and $\ell_1 + \ell_2 + \ell$ odd forces at least one pair to be equal in this range). By Proposition 14, all odd-parity entries vanish. Therefore, the T_R ambiguity cannot be resolved using only degrees ≤ 3.

Remark 18 (Why the augmented block is needed). The bootstrap block $\mathcal{T}_\ell$ alone is rank-deficient on the real-signal slice: at the seed-42 real witness, the $(2\ell+1) \times (2\ell+1)$ submatrix A_ℓ assembled from the $\mathcal{T}_\ell$ rows has SVD rank $\approx \ell+2$ rather than $2\ell+1$ for $\ell \in \{8, \ldots, 15\}$ (verified numerically; the deficit is generic over real witnesses). The mandatory even self-couplings and CG-power scalars are quadratic or cubic in $\mathbf{F}_\ell$ and supply the additional independent real equations needed to fill out the per-degree real Jacobian; the augmented per-degree block (linear $\mathcal{T}_\ell$ plus quadratic self-couplings plus CG-power scalars) is what Algorithm 2 assembles, and is what the implementation actually solves at degrees $\ell \geq 5$.

A.5 Comparison with existing results

Table 9: Comparison of bispectral invariants for signals on S^2 or finite groups.

Result	Invariant	Size	Complete?
Kakarala [Kakarala, 2012]	Full bispec.	$O(L^3)$	Yes (constr.)
Edidin & Satriano [Edidin and Satriano, 2024]	Full bispec. (band-lim.)	$O(L^3)$	Yes (generic)
Bendory et al. [Bendory et al., 2025]	Freq. marching	$O(L^2)$/shell	Yes (3 shells)
Mataigne et al. [Mataigne et al., 2024]	Sel. G-bispec. (finite)	$O(\|G\|)$	Yes (exact)
This work	**Sel.** $\mathrm{SO}(3)$ **on** S^2	$\boldsymbol{\Theta(L^2)}$	**Empirical (Section 6.2)**

A.6 Complexity in pixel space

On S^2, a band-limited signal at degree L is sampled on an equiangular grid of size $N_\theta \times N_\varphi \approx 2L \times 2L$, so $P = 4L^2$ and $L = \Theta(\sqrt{P})$.

Output size.

Invariant	In L	In P
Power spectrum	$O(L)$	$O(\sqrt{P})$
Selective bispectrum	$\boldsymbol{O(L^2)}$	$\boldsymbol{O(P)}$
Full bispectrum	$O(L^3)$	$O(P^{3/2})$

The selective bispectrum is *linear* in the number of pixels.

Computation cost.

Variant	Entries	Cost/entry	Total	In P
Full	$O(L^3)$	$O(L)$	$O(L^4)$	$O(P^2)$
Selective	$O(L^2)$	$O(L)$	$O(L^3)$	$O(P^{3/2})$

Concrete example. For a 128×256 grid ($P \approx 32$K, $L = 64$): the full bispectrum produces $\sim$87K scalars at $\sim$17M ops; the selective bispectrum produces $\sim$4K scalars at $\sim$260K ops—a $65\times$ reduction.

B Extended Benchmark Results

This appendix supplements the summary in Table 3 (main text) with scaling plots produced by `benchmarks/benchmark.py` on a single NVIDIA H100 80 GB GPU (PyTorch 2.10).

B.1 Coefficient-count scaling

B.2 Forward-pass timing

The selective forward pass is dominated by kernel-launch overhead rather than arithmetic, making it effectively free relative to backbone computation. The full bispectrum exhibits the expected super-linear growth and becomes impractical beyond $|G| \approx 256$ for C_n.

B.3 GPU throughput and inversion

All modules exhibit linear throughput scaling with batch size, confirming that the implementation is memory-bandwidth-bound rather than compute-bound at these scales.

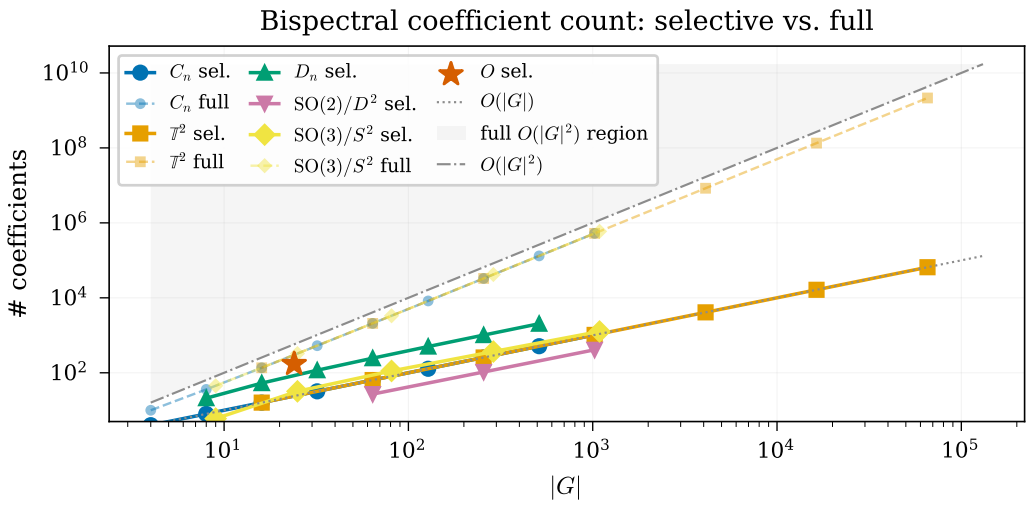

Figure 4: Bispectral coefficient count vs. group order $|G|$. Solid lines: selective ($O(|G|)$); dashed: full ($O(|G|^2)$). At $|G|$=1,024, selectivity yields a 512× reduction. Continuous groups (SO(2), SO(3)) are plotted at their band-limited signal dimension; full-bispectrum curves for D_n and the disk are omitted (backends not yet implemented).

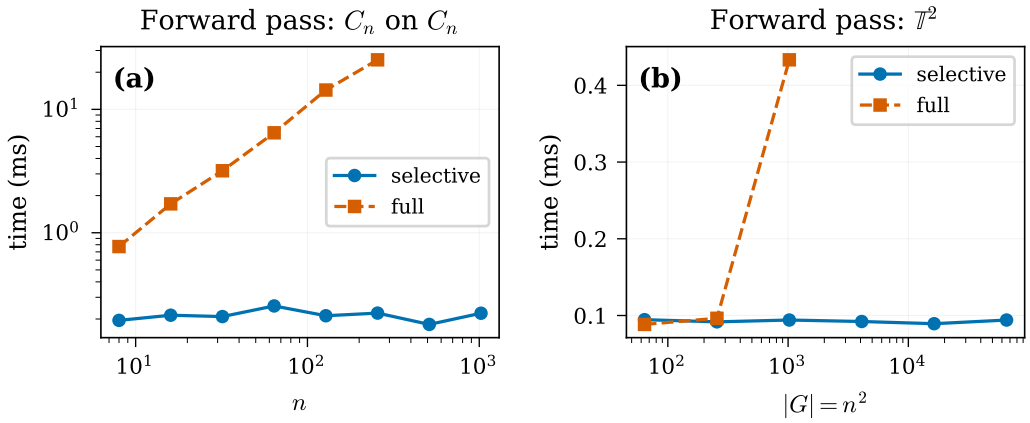

Figure 5: Forward-pass wall-clock time (batch = 16, H100 GPU). **(a)** C_n: the selective pass stays flat at ∼0.14 ms regardless of n, while the full bispectrum grows super-linearly (17 ms at n=256). **(b)** $\mathbb{T}^2$: selective is constant at ∼0.08 ms up to $|G|$=65,536.

C PCam: Extended Results

C.1 Published baselines

Table 10 places our results in the context of published methods on PCam and the closely related Camelyon16 benchmark. Our models operate at 68K–920K parameters and are trained on only 10% of the data.

C.2 Model variants

All equivariant models share a C_8-equivariant DenseNet [Huang et al., 2017] backbone with block configuration $(4, 4, 4)$, 24 initial channels, and compression factor 0.5, following the architecture of Veeling et al. [2018] but replacing the final invariant map. Six model variants are evaluated:

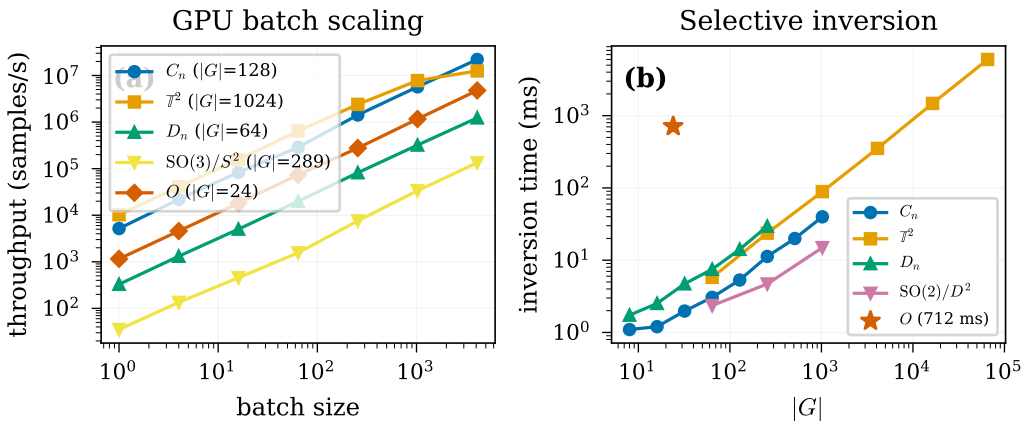

Figure 6: **(a)** GPU throughput (samples/s) vs. batch size for the selective forward pass. C_n and $\mathbb{T}^2$ exceed 10^7 samples/s; the octahedral module reaches 5.7×10^6 samples/s. All modules scale linearly with batch size. **(b)** Inversion wall-clock time. Abelian groups (C_n, D_n, SO(2)/disk) scale roughly linearly. SO(3) on S^2 inversion is not yet implemented.

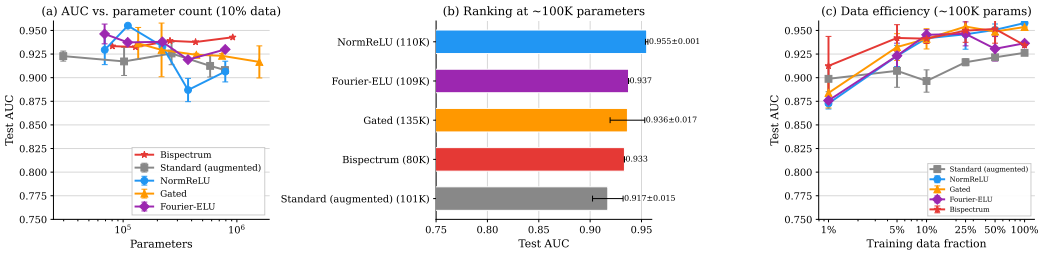

Figure 7: PCam Pareto analysis (`CnonCn` selective bispectrum, single seed, 10% training data). (a) Test AUC vs. parameter count across all five pooling variants. (b) Ranking at the ∼100K parameter budget. (c) Data efficiency at matched ∼100K parameters. Multi-seed statistics at matched capacity are in Table 4 (main text).

- **Standard**: vanilla CNN with random rotation/reflection augmentation and global average pooling. No group dimension, so ∼8× fewer parameters at matched growth rate.
- **NormReLU**: C_8-equivariant backbone with norm-based nonlinearity and max pooling over the group dimension.
- **Gated**: C_8-equivariant backbone with gated nonlinearity (doubles internal channels) and group max pool.
- **Fourier-ELU**: C_8-equivariant backbone with Fourier-space ELU nonlinearity and group max pool.
- **Bispectrum**: C_8-equivariant backbone with ReLU nonlinearity and selective G-bispectrum via `CnonCn`(n=8) as the terminal invariant map.

C.3 Training details

To enable a controlled Pareto comparison, we sweep the DenseNet growth rate $k \in \{3, 4, 6, 8, 12\}$ for equivariant models (and $k \in \{6, 12, 20, 30, 35\}$ for the standard CNN to cover a comparable parameter range), yielding models from ∼30K to ∼1.6M parameters. All models are trained on **10% of training data** (26,214 images) with AdamW [Loshchilov and Hutter, 2019] (lr $= 10^{-3}$, weight decay 10^{-4}), cosine warm restarts, mixed-precision training, and early stopping on validation AUC (patience 10, max 50 epochs).

Table 10: Published baselines on PatchCamelyon from Veeling et al. [2018], trained on 100% data.

Method	Reference	Params	ACC	AUC
P4M-DenseNet (C_8)	Veeling et al. 2018	119K	0.898	0.963
P4M-DenseNet M	Veeling et al. 2018	19K	0.893	0.958
DenseNet (augmented)	Veeling et al. 2018	128K	0.881	0.951
DenseNet (no aug.)	Veeling et al. 2018	128K	0.876	0.955

C.4 Pareto analysis interpretation

Architectural stability (single seed, growth-rate sweep). The bispectrum's AUC varies by only $\sim$1pp across a $12\times$ range of model sizes (80K–920K parameters), while norm swings by $>$7pp. For practitioners, bispectral pooling "just works" without growth rate tuning. The multi-seed matched-parameter results (Table 4) confirm this: bispectrum and Fourier-ELU have the lowest cross-seed variance (± 0.004).

Matched parameters. At matched parameters ($\sim$100K), all equivariant models outperform the standard CNN (AUC 0.896 ± 0.010, 3 seeds). Fourier-ELU leads slightly (0.945 ± 0.004), followed by NormReLU (0.942), bispectrum and gated (both 0.941). All equivariant methods cluster within $\sim$0.5pp of each other.

Scaling behavior. As parameters increase from the smallest to the largest configuration, standard, norm, and Fourier-ELU all degrade (by 1.8, 2.6, and 0.9 AUC points respectively), while the bispectrum improves (+0.9 points, reaching 0.943 at 920K params). This suggests that the bispectrum's complete invariant map scales more gracefully with capacity.

Data scaling (3 seeds). At 1% data the bispectrum leads all methods (0.912 ± 0.025), outperforming NormReLU (0.873), Fourier-ELU (0.876), and gate (0.884). By 10% all equivariant methods converge ($\sim$0.941–0.945). The bispectrum's data-efficiency advantage narrows as data increases, consistent with the hypothesis that complete invariants substitute for missing training examples in the low-data regime.

D OrganMNIST3D: Extended Results

D.1 Published baselines

Table 11 situates our results against published methods on the same benchmark. Our models are deliberately small ($\leq$463K parameters) to enable controlled ablation of the pooling mechanism. The relevant comparison is against equivariant methods at a similar scale.

Table 11: Published baselines on OrganMNIST3D for context. Accuracy differences with respect to large-scale models reflect the deliberate parameter constraint of our ablation study.

Method	Reference	Params	ACC	AUC
ResNet-18 + 3D	Yang et al. 2023	33M	0.907	0.996
ResNet-18 + ACS	Yang et al. 2023	11M	0.900	0.994
ILPOResNet-50	Zhemchuzhnikov and Grudinin [2024]	38K	0.879	0.992
EquiLoPO (local train.)	Zhemchuzhnikov and Grudinin [2025]	418K	0.866	0.991
SE3MovFrNet	Sangalli et al. [2023]	—	0.745	—
Regular SE(3) conv	Kuipers and Bekkers [2023]	172K	0.698	—

D.2 Rotation analysis

Figure 8 shows that all equivariant models achieve identical performance on the original and octahedrally rotated test sets ($\sigma_{\mathrm{rot}} \approx 0$), while the augmentation-trained standard CNN degrades under rotation.

D.3 Wider channels

To test whether the advantage of bispectral completeness persists at higher capacity, we run max pool and bispectrum at channel widths $(8, 16)$ and $(16, 32)$ with full training data.

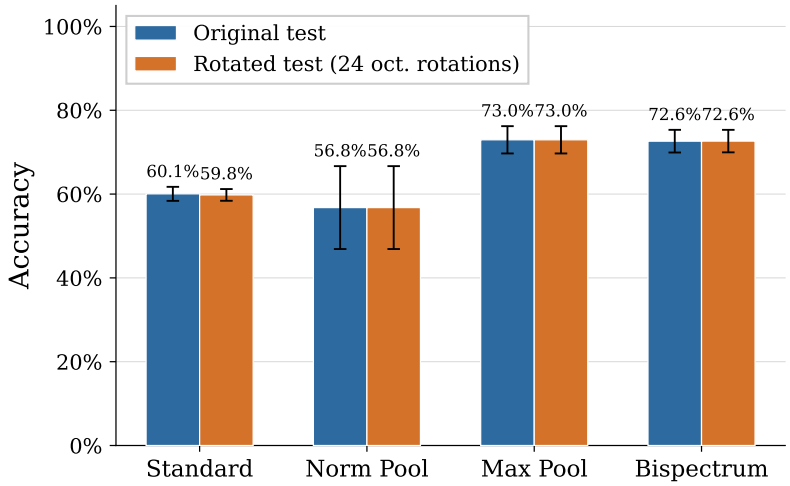

Figure 8: OrganMNIST3D test accuracy on the original and octahedrally rotated test sets. Error bars show standard deviation over three seeds. All equivariant models achieve identical performance under rotation ($\sigma_{\text{rot}} \approx 0$), while the augmentation-trained standard CNN degrades.

Table 12: OrganMNIST3D accuracy at wider channel configurations (mean $\pm$ std over 3 seeds, full training data).

Channels	Params	Max Pool	Bispectrum
$(4, 8)$	$\sim$375K / 463K	$.730 \pm .033$	$.726 \pm .027$
$(8, 16)$	$\sim$1.5M / 1.7M	$.743 \pm .015$	$.745 \pm .006$
$(16, 32)$	$\sim$6.0M / 6.3M	$\mathbf{.785 \pm .032}$	$.685 \pm .039$

At $(8, 16)$ the two methods are equivalent ($74.5 \pm 0.6\%$ bispectrum vs. $74.3 \pm 1.5\%$ max pool, both at $\sim$1.5–1.7M parameters). At $(16, 32)$ max pool reaches $78.5 \pm 3.2\%$ while bispectrum drops to $68.5 \pm 3.9\%$—the 6.3M parameter bispectrum model likely overfits the 971 training samples. This suggests that the bispectrum's completeness advantage is most pronounced in the data-limited rather than the parameter-limited regime, consistent with the data efficiency findings in the main text.

E Experimental Details

E.1 Hyperparameters

Table 13 summarises the training hyperparameters for all three experimental settings.

Table 13: Training hyperparameters across experiments.

	PCam	OrganMNIST3D	Sph. MNIST
Optimizer	AdamW	AdamW	AdamW
Learning rate	10^{-3}	10^{-3}	10^{-3}
Weight decay	10^{-4}	10^{-4}	10^{-4}
Batch size	64–256	16–64	256
Scheduler	Cosine warm restarts	—	—
Max epochs	50	100	50
Early stopping	Val AUC, pat. 10	Val AUC, pat. 15	Val ACC, pat. 10
Seeds	42, 123, 456†	42, 123, 456	42, 123, 456
Mixed precision	Yes	No	No

†Pareto figure uses seed 42; matched-parameter table and data efficiency use seeds 42, 123, 456.

E.2 Hardware

All experiments were run on a single NVIDIA H100 GPU (80 GB). Training wall-clock times range from $\sim$5 minutes (OrganMNIST3D, 971 samples) to $\sim$20 minutes (PCam, 26K samples at 10%

data). Spherical MNIST bispectrum computation (forward pass through `SO3onS2` at $\ell_{\max} = 15$) takes ∼0.3s per batch of 64 on GPU.

E.3 Data preprocessing

PCam. PatchCamelyon [Veeling et al., 2018] consists of 327,680 RGB patches of size 96×96 pixels extracted from whole-slide images of lymph node sections [Ehteshami Bejnordi et al., 2017]. We use the standard train/val/test split. Images are loaded from HDF5 files and normalised to $[0, 1]$. The standard (non-equivariant) baseline uses random rotation and reflection augmentation during training; all equivariant models are trained without augmentation.

OrganMNIST3D. OrganMNIST3D [Yang et al., 2023] contains 1,742 abdominal CT organ volumes of size $28 \times 28 \times 28$ voxels with 11 organ classes (train/val/test = 971/161/610). Volumes are normalised to $[0, 1]$. The standard baseline uses random octahedral augmentation (one of 24 rotations per sample); equivariant models are trained without augmentation. Kernel rotations are implemented as exact voxel permutations on the 3^3 grid, avoiding interpolation artifacts.

Spherical MNIST. MNIST digits are projected onto the sphere via stereographic projection following Cohen et al. [2018] and sampled on a 64×128 equiangular grid. The selective bispectrum is computed at $\ell_{\max} = 15$, producing 384 complex invariant coefficients. The model input is obtained by applying $\mathrm{sign}(x)\log(1 + |x|)$ separately to the real and imaginary parts of each coefficient (a sign-preserving, per-channel injective transform), giving 768 real features. These features are fed to a 3-layer MLP: Linear(768, 256)–BN–ReLU–Linear(256, 128)–BN–ReLU–Linear(128, 10). The power spectrum MLP uses the same structure with hidden=128: Linear(16, 128)–BN–ReLU–Linear(128, 64)–BN–ReLU–Linear(64, 10). The standard CNN has 4 convolutional layers (16–32–64–64 channels) followed by adaptive average pooling and a Linear(1024, 128)–ReLU–Linear(128, 10) classifier.

E.4 Aggregation procedures

PCam. The Pareto figure (Figure 7) uses a single seed (42) across the full growth-rate sweep. The matched-parameter table (Table 4) and data efficiency analysis use 3 seeds (42, 123, 456) per configuration.

OrganMNIST3D. All results in Table 5 report mean ± std over 3 seeds (42, 123, 456). Rotation robustness (σ_{rot}) is the standard deviation of test accuracy across all 24 octahedral rotations of the test set. The data efficiency analysis (Figure 1 in the main text) uses 3 seeds per data fraction.

Spherical MNIST. All results in Table 6 report mean ± std over 3 seeds (42, 123, 456). Rotation robustness (σ_{rot}) is measured by applying 10 random $\mathrm{SO}(3)$ rotations to the entire test set and computing the standard deviation of accuracy, yielding 30 measurements per model (10 rotations × 3 seeds).

E.5 Spherical MNIST: bispectrum reconstruction protocol

This section gives the protocol behind Figure 3 (Section 6.2). The implementation lives in `paper/experiments/spherical_mnist_reconstruction/reconstruct.py`.

Setup. Spherical MNIST is rendered onto an equiangular $64{\times}128$ lat–lon grid using the same stereographic projection as the classifier, then band-limited at $L = 12$ via a forward/inverse SHT roundtrip (`torch_harmonics.RealSHT`). For each digit we draw a target rotation $R \in \mathrm{SO}(3)$ uniformly at random and form $f := R \cdot f_0$. We compute the SO(3)-on-S^2 selective bispectrum $\beta := \Phi_{\mathrm{sel}}(f)$.

Reconstruction. Starting from a Gaussian initialization $\hat{f}_0 \sim \mathcal{N}(0, I)$ on the $64{\times}128$ grid, we optimize $\hat{f}$ with Adam (initial LR 5×10^{-2}, cosine annealing to 5×10^{-4}, 8000 steps) to minimize the relative complex L2 loss $\|\beta(\hat{f}) - \beta\|^2/\|\beta\|^2$. After every step we project $\hat{f}$ back to the band-limited subspace via $\mathrm{SHT}^{-1} \circ \mathrm{SHT}$ truncated at L to suppress the unconstrained null space. We

run 4 random restarts and keep the $\hat{f}$ with the lowest bispectrum residual. Reconstruction typically reaches $\|\beta(\hat{f}) - \beta\|/\|\beta\| \sim 10^{-3}$, near the SHT-discretization invariance floor for this grid.

$\mathrm{SO}(3)$ **alignment.** The bispectrum is invariant on $\mathrm{SO}(3)$-orbits, so $\hat{f}$ lives somewhere in the orbit of f rather than at f itself. We resolve the orbit ambiguity by Procrustes-style fitting: we parameterize $\hat{R} \in \mathrm{SO}(3)$ as a unit quaternion (auto-normalized) to avoid gimbal lock, and minimize $\|\hat{R} \cdot \hat{f} - f\|^2$ with Adam (12 random restarts, 200 steps each, cosine-annealed LR). Rotations of the spherical image use bilinear interpolation in lat–lon coordinates (`bispectrum.rotate_spherical_function`).

Numerical floor. On a 64×128 grid, two SHT roundtrips of two $\mathrm{SO}(3)$-rotated copies of the same signal already differ by $\|f' - f\|/\|f\| \approx 2 \times 10^{-2}$ in image space due to bilinear-interpolation discretization; this is the noise floor below which alignment cannot improve. The aligned residuals reported in Figure 3 (0.1–0.3) sit one to two orders of magnitude above this floor and are dominated by gradient descent local optima on the bispectrum reconstruction step, not by missing information.

E.6 Dataset licenses

- **PCam**: CC0 1.0 Universal (public domain).
- **OrganMNIST3D**: CC BY 4.0.
- **Spherical MNIST**: derived from MNIST, which is publicly available for research use.

F API Example

```
import torch
from bispectrum import CnonCn, SO2onDisk, OctaonOcta

bsp = CnonCn(n=8, selective=True)
f = torch.randn(16, 8)
beta = bsp(f)
f_rec = bsp.invert(beta)
```

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Algorithm 2 Augmented selective index set for the $\mathrm{SO}(3)$ bispectrum on S^2 at band-limit L. Returns bispectral triples $\mathcal{S}_\beta$ and CG power triples $\mathcal{S}_P$. This extends the BFS-based selective index construction of Mataigne et al. [2024] from finite groups to $\mathrm{SO}(3)$ on S^2, adding the CG power augmentation step that is specific to real signals.

Input: Band-limit L
Output: Bispectral set $\mathcal{S}_\beta$, CG power set $\mathcal{S}_P$
$\mathcal{S}_\beta \leftarrow \{(0,0,0)\}$; $\mathcal{S}_P \leftarrow \emptyset$
Seed block: add to $\mathcal{S}_\beta$ all triples (ℓ_1, ℓ_2, ℓ) with $0 \le \ell_1 \le \ell_2 \le L_0 = 4$ and $|\ell_1 - \ell_2| \le \ell \le \min(\ell_1 + \ell_2, L_0)$ {constant-size full-bispectrum seed}
for $\ell = 1, \ldots, L$ **do**
  budget $\leftarrow 2\ell + 1$
  if $\ell = 4$ **then**
    budget $\leftarrow 10$ {keep all chain+cross at $\ell = 4$}
  end if
  candidates $\leftarrow [\,]$
  if $4 \le \ell \le 7$ **then**
    append the explicit low-degree linear block $\mathcal{T}_\ell^{\mathrm{small}}$ to candidates
  else if $\ell \ge 8$ **then**
    for $a = 1, \ldots, \ell{-}1$ **do**
      append $(a,\ \ell,\ \ell{-}a)$ to candidates
    end for
    for $a = 2, \ldots, \ell{-}1$ **do**
      if not (ℓ is odd **and** $a = (\ell{+}1)/2$) **then**
        append $(a,\ \ell,\ \ell{-}a{+}1)$ to candidates {skip $(r, 2r{-}1, r) \equiv 0$ on real signals, Prop. 14}
      end if
    end for
    for $a = 1, \ldots, 4$ **do**
      append $(a,\ \ell{-}a,\ \ell)$ to candidates
    end for
    if ℓ is odd **then**
      append $(2,\ \ell{-}1,\ \ell)$ to candidates {compensate excluded $(r, 2r{-}1, r)$ row}
    end if
  else
    append low-degree chain candidates
    if $\ell \ne 2$ **then**
      append cross candidates {$\beta_{1,2,1}$ excluded}
    end if
  end if
  append $(0,\ \ell,\ \ell)$ to candidates {power entry}
  if $\ell \notin \{1,\ 2\}$ **then**
    for $\ell' = \ell{-}1, \ldots, 0$ **do**
      append $(\ell,\ \ell,\ \ell')$ to candidates
    end for
  end if
  if $\ell = 2$ **then**
    append $(\ell,\ \ell,\ \ell)$ to candidates $\{\beta_{2,2,2}\}$
  end if
  Remove entries that vanish for real signals; deduplicate against $\mathcal{S}_\beta$ already added by the seed block
  $\mathcal{S}_\beta \leftarrow \mathcal{S}_\beta \cup$ first $\min(\text{budget},\ |\text{candidates}|)$ live entries
  {**Mandatory even self-couplings**}
  for $\ell' = 2, 4, \ldots, \lfloor \ell/2 \rfloor \cdot 2$ with $\ell' \le \ell$ and $\ell \ge 3$ **do**
    $\mathcal{S}_\beta \leftarrow \mathcal{S}_\beta \cup \{(\ell,\ \ell,\ \ell')\}$
  end for
  {**CG power augmentation**}
  Greedily add $P_{\ell_1,\ell_2,\ell'}$ (with $\max(\ell_1, \ell_2) = \ell$, $|\ell_1{-}\ell_2| \le \ell' \le \ell_1{+}\ell_2$) to $\mathcal{S}_P$ until the Jacobian rank of the per-degree restriction $\Phi_{\mathrm{sel}}|_\ell$ matches the number of unknowns
end for
return $(\mathcal{S}_\beta,\ \mathcal{S}_P)$